UNIVERSITÀ DI PARMA

Dipartimento di Ingegneria e Architettura

Corso di

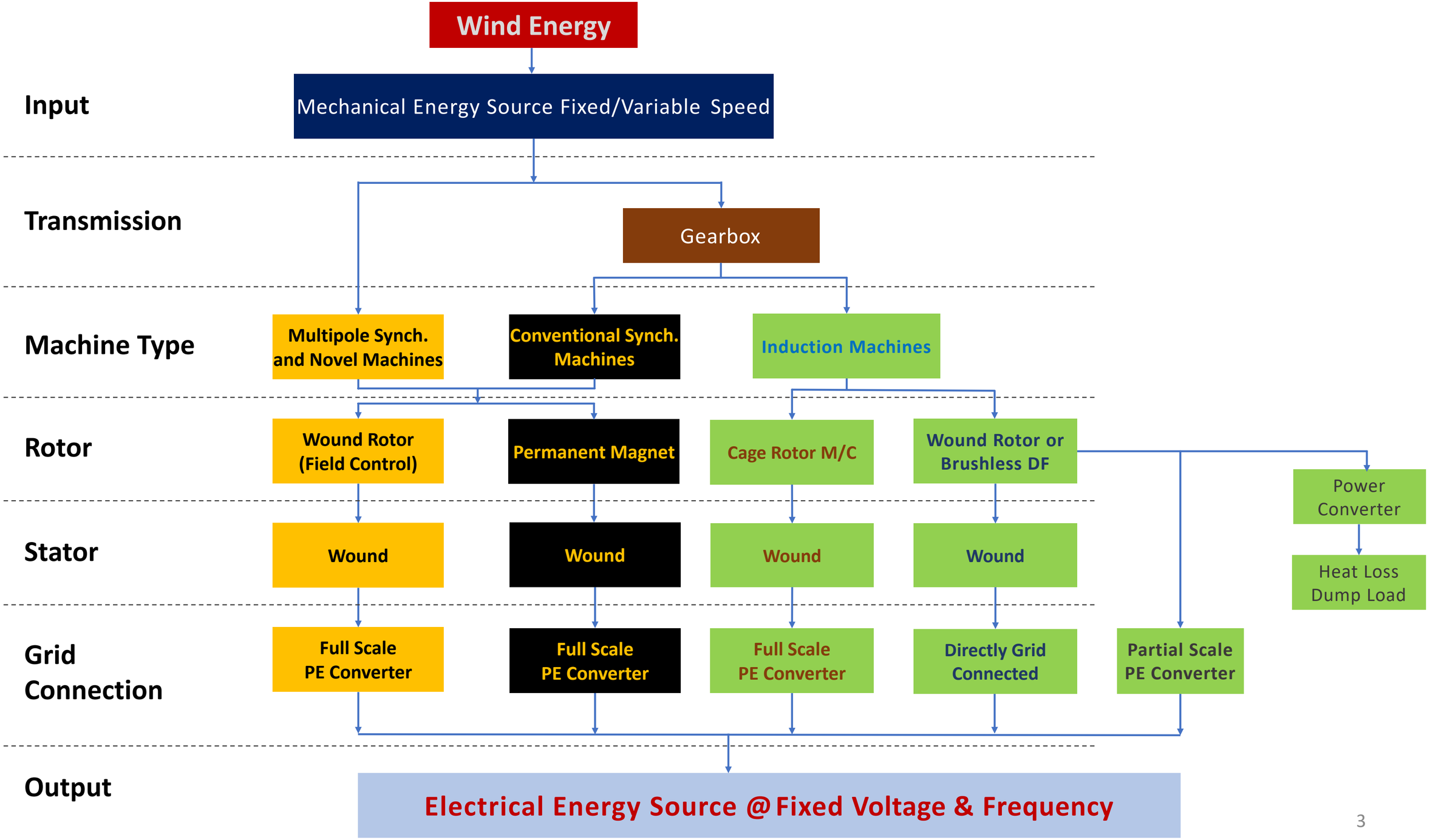
Elettronica per la conversione dell'energia e le fonti rinnovabili

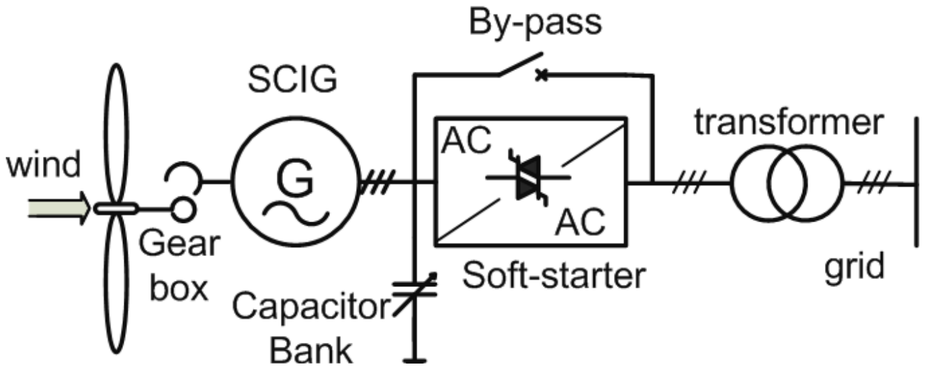
Sistemi Eolici
Parte 2 di 2

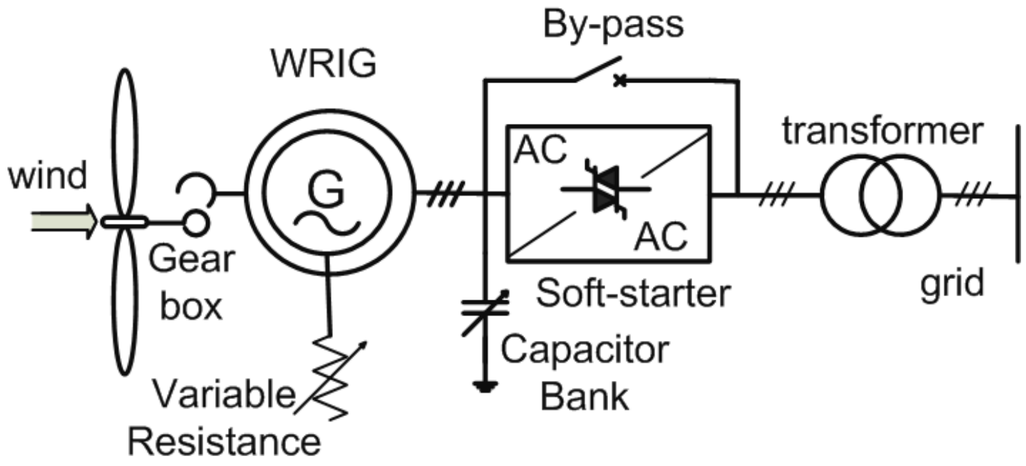
Nicola Delmonte

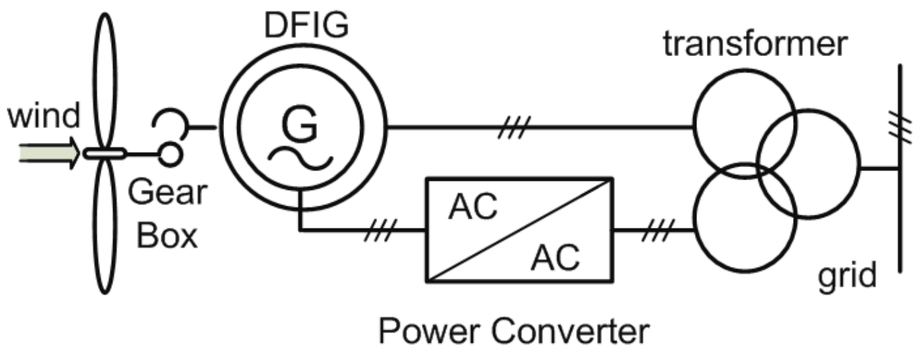
Sistemi eolici (2^ parte)

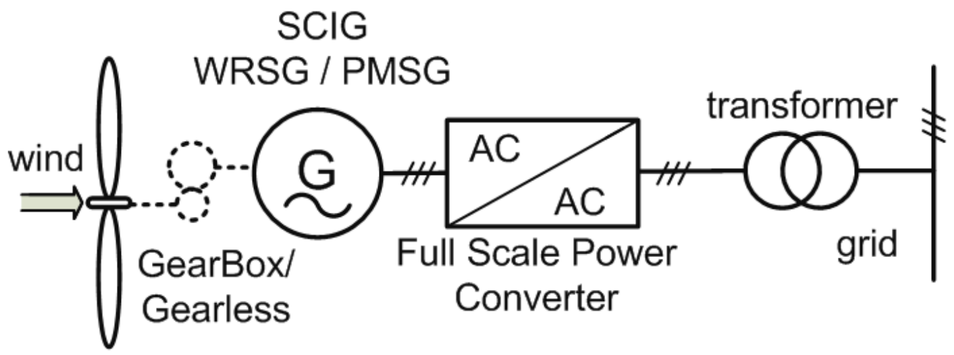
- Cascata di conversione: alternative
- Topologie dei convertitori di potenza nei sistemi eolici
- Convertitori multlivello


Wind Energy
Input
Mechanical Energy Source Fixed/Variable Speed
Transmission
Gearbox
Machine Type
Multipole Synch. and Novel Machines
Conventional Synch. Machines
Induction Machines
Rotor
Wound Rotor (Field Control)
Permanent Magnet
Cage Rotor M/C
Wound Rotor or Brushless DF
Power Converter
Heat Loss Dump Load
Stator
Wound
Wound
Wound
Wound
Grid Connection
Full Scale PE Converter
Full Scale PE Converter
Full Scale PE Converter
Directly Grid Connected
Partial Scale PE Converter
Output
Electrical Energy Source @ Fixed Voltage & Frequency

Fixed Speed with SCIG	
Control Active power	Limited
Control Reactive Power	No
Short circuit (fault-active)	No
Short circuit power	contribute
Control BW	1-10s
Standby function	No
Softstard needed	Yes
Reactive compensator (C)	Yes
Island operation	No
Investment	++
Maintenance	++

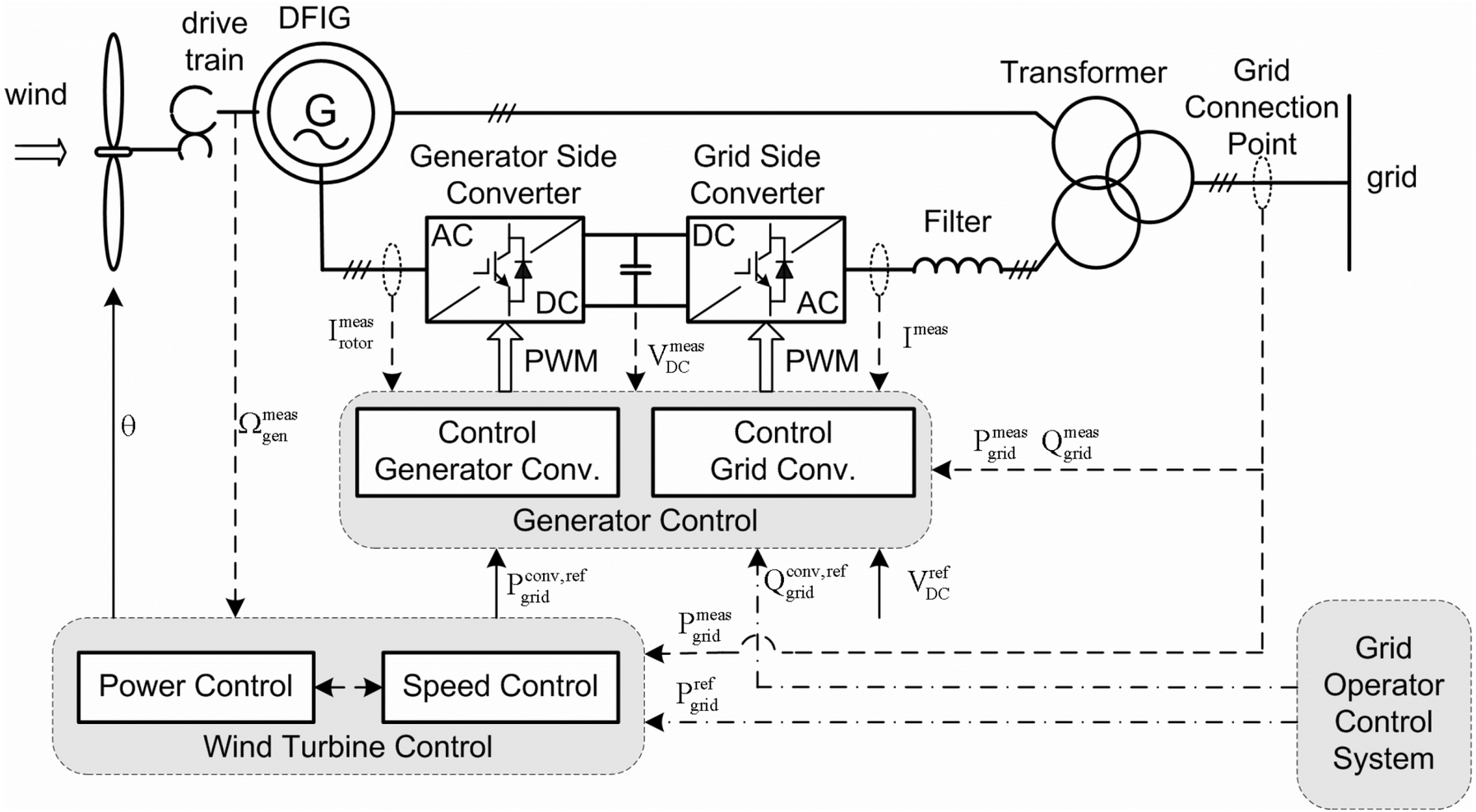
Fixed Speed with WRIG	
Control Active power	Limited
Control Reactive Power	No
Short circuit (fault-active)	No
Short circuit power	contribute
Control BW	100ms
Standby function	No
Softstard needed	Yes
Reactive compensator (C)	Yes
Island operation	No
Investment	++
Maintenance	++

Variable Speed with DFIG	
Control Active power	Yes
Control Reactive Power	Yes
Short circuit (fault-active)	No/Yes
Short circuit power	Contribute
Control BW	1ms
Standby function	Yes+
Softstard needed	No
Reactive compensator (C)	No
Island operation	Yes/No
Investment	+
Maintenance	=

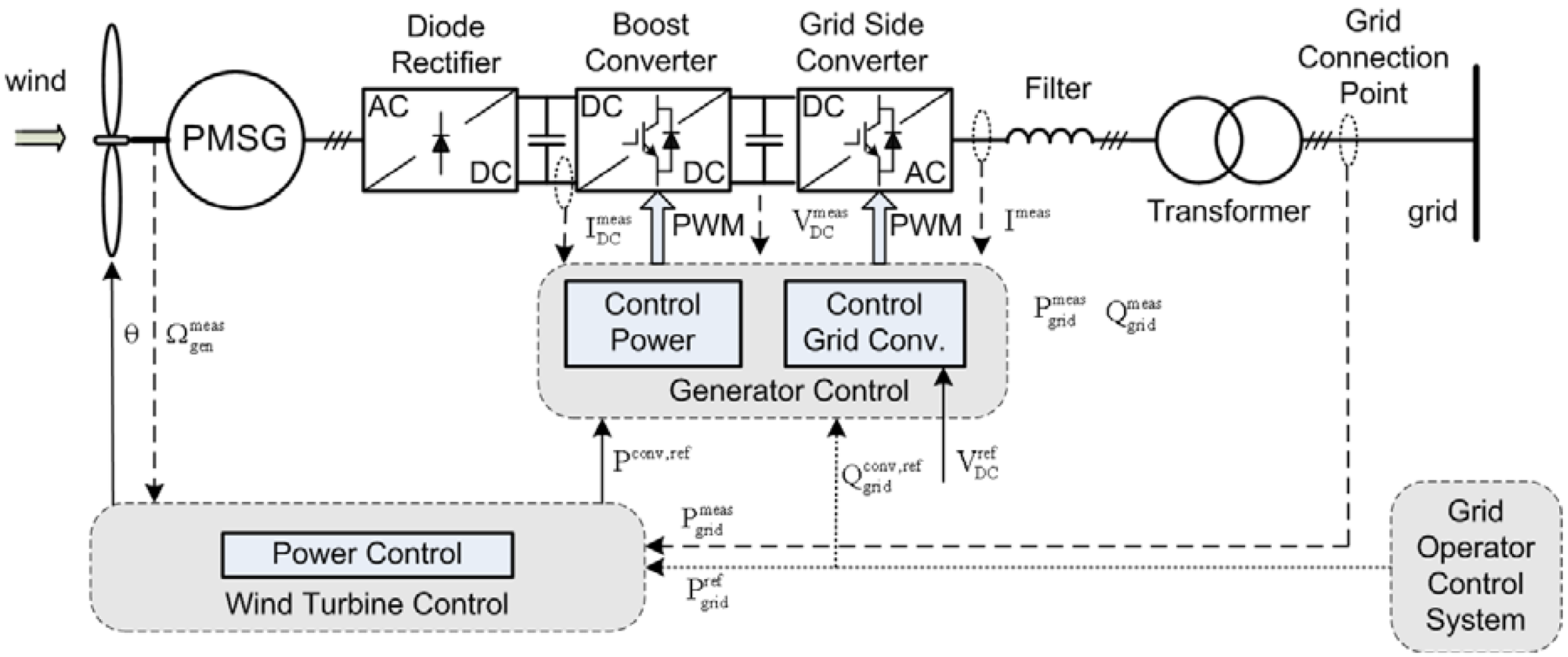
Variable Speed with Full-Scale converter	
Control Active power	Yes
Control Reactive Power	Yes
Short circuit (fault-active)	Yes
Short circuit power	Limit
Control BW	0.5-1ms
Standby function	Yes++
Softstard needed	No
Reactive compensator (C)	No
Island operation	Yes
Investment	=
Maintenance	+

Power System comparison of Wind Turbine configurations

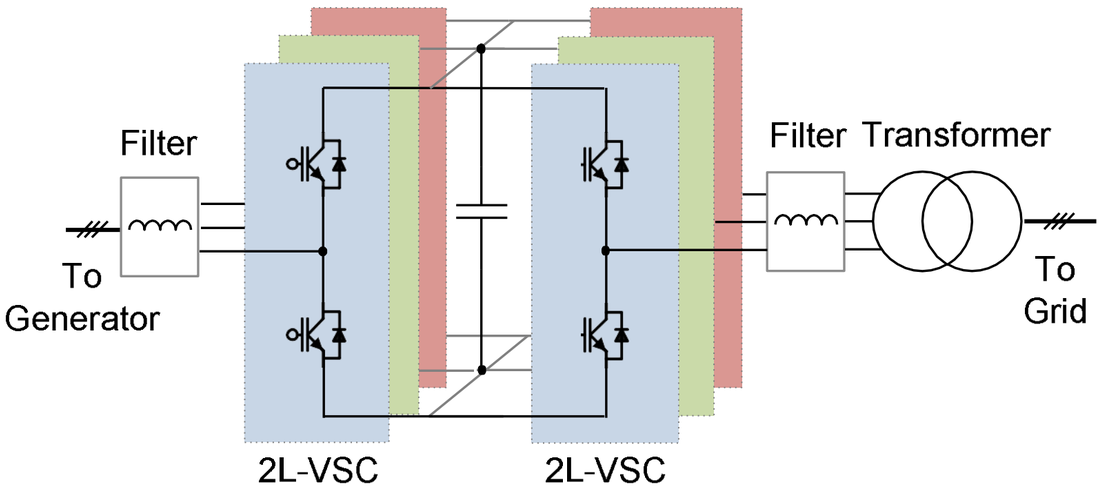
Configuration	2L BTB	3L-NPC BTB	3L-HB BTB	5L-HB BTB	3L-NPC + 5L-HB	CHB-MFT
Output Voltage Level	2	3	3	5	3/5	2N+1
IGBT number	1pu	2ui	2pu	4pu	3pu	4N pu
Diode number	1pu	3pu	2pu	6pu	4/5pu	4N pu
Max Output Voltage	1pu	2ui	2pu	4pu	2/4pu	2N pu
Fault Tolerant Ability	No	No	Limited	Limited	No	Yes
Filter Size	+++	++	++	+	++/+	=
Tansformer Size	+++	+++	+++	+++	+++	+



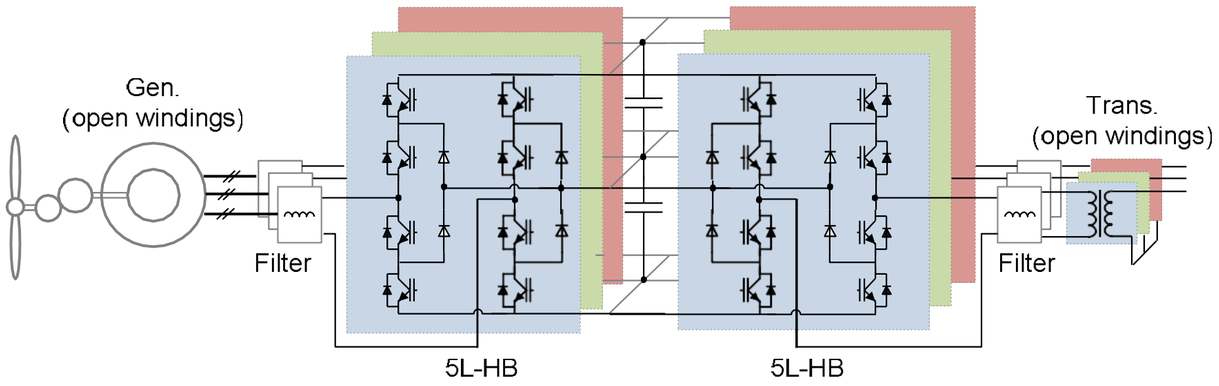
Control of a wind turbine with doubly-fed induction generator



Control of a wind turbine with multi-pole synchronous generator



Two-level back-to-back voltage source converter for wind turbines. (**2L BTB**)



Five-level H-bridge back-to-back converter for wind turbines. (**5L-HB BTB**)

Conversione multilivello

Necessità:

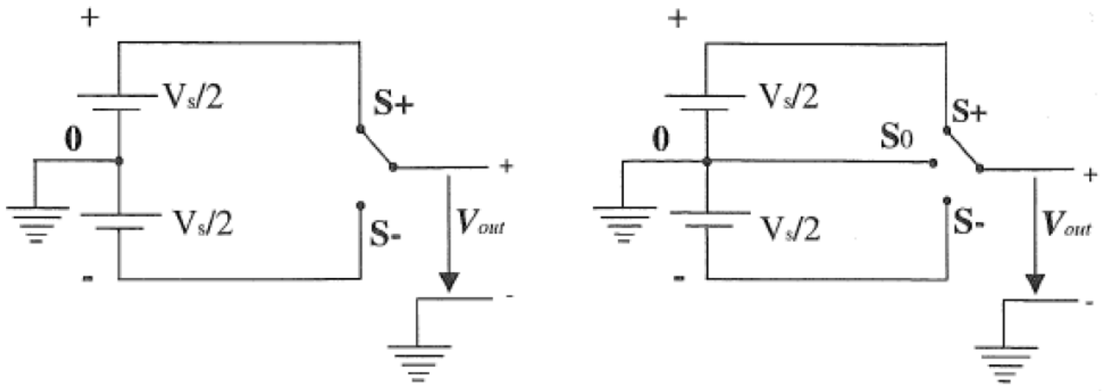
- Potenze da erogare sempre più elevate con aumento dell'efficienza del convertitore
- AC a 50 Hz prodotta con la minor THD possibile

Desiderio:

- Impiegare frequenze di switching sempre più elevate per ridurre il volume dei componenti magnetici

Possibile soluzione topologica:

- Inverter multilevel



a) Two-level b) Three-level

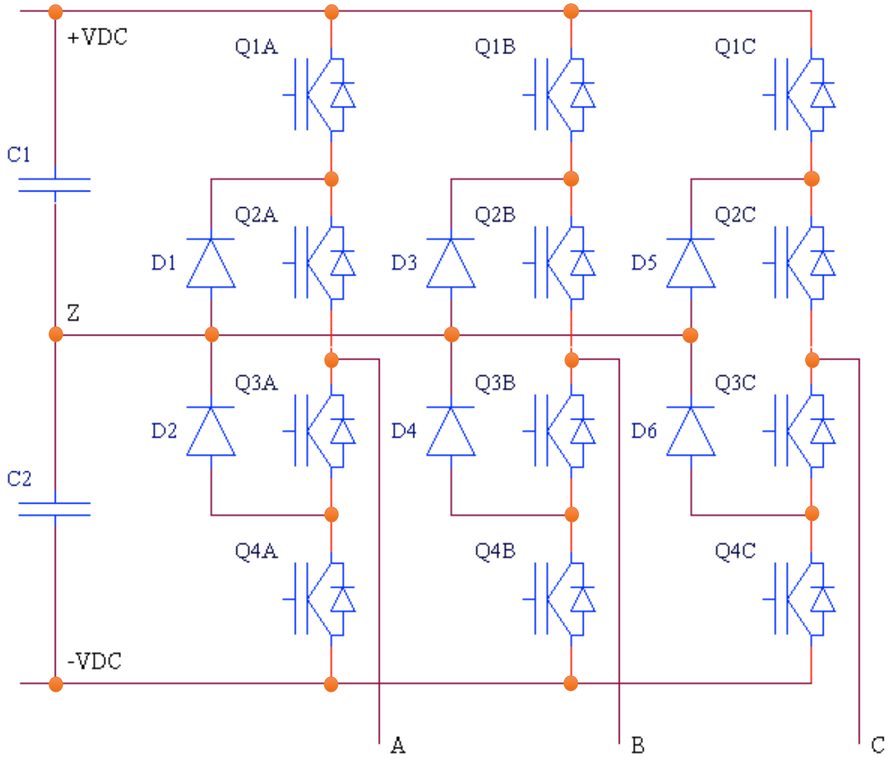
Questi presentano numerosi vantaggi rispetto agli inverter due livelli: forniscono all'uscita forme d'onda a gradini su tre o più livelli di tensione con conseguente miglioramento dello spettro armonico; possono operare a tensioni e potenze molto elevate; possono limitare la frequenza di lavoro degli interruttori.

Come sorgenti di tensione è impiegata una serie di condensatori, il cui numero dipende dal numero di livelli dell'inverter. In un inverter monofase ad *n* livelli il bus DC di ingresso necessita di *n*-1 condensatori. Le varie combinazioni degli interruttori permettono di sommare in uscita le varie tensioni dei condensatori.

Conversione multilivello

Esistono diverse topologie che permettono di ottenere una conversione a più livelli. Tre di queste, fra quelle più diffuse, sono:

- *Diode clamped*
- *Capacitor clamped*
- Cascata di convertitori a 2 livelli

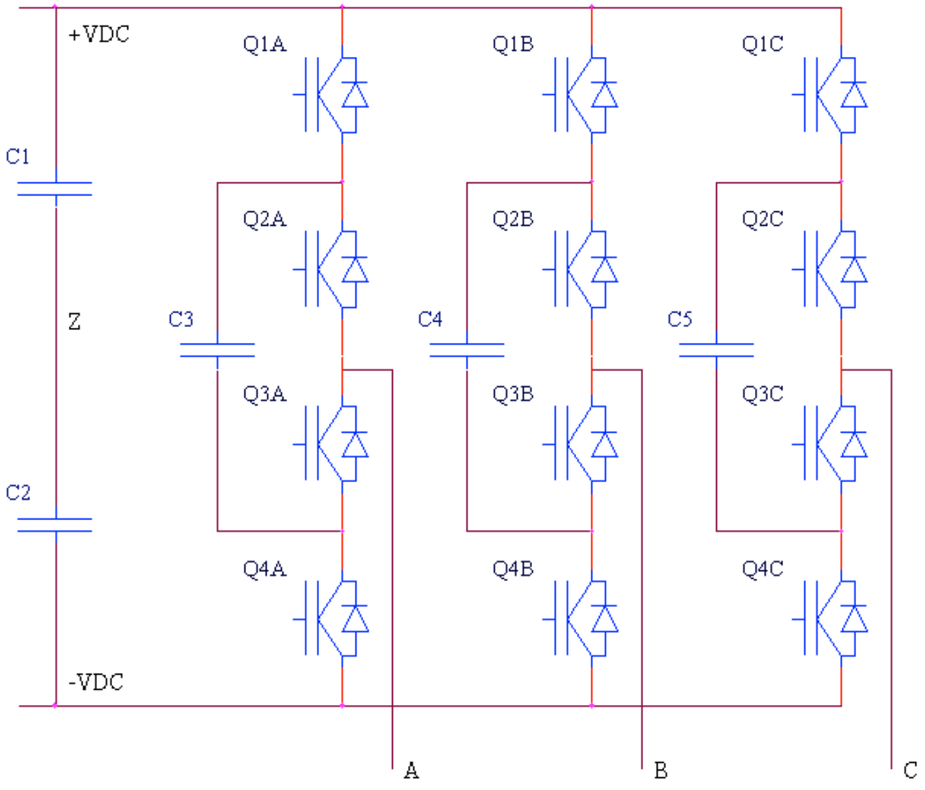


3-phase clamped diode DC/AC converter

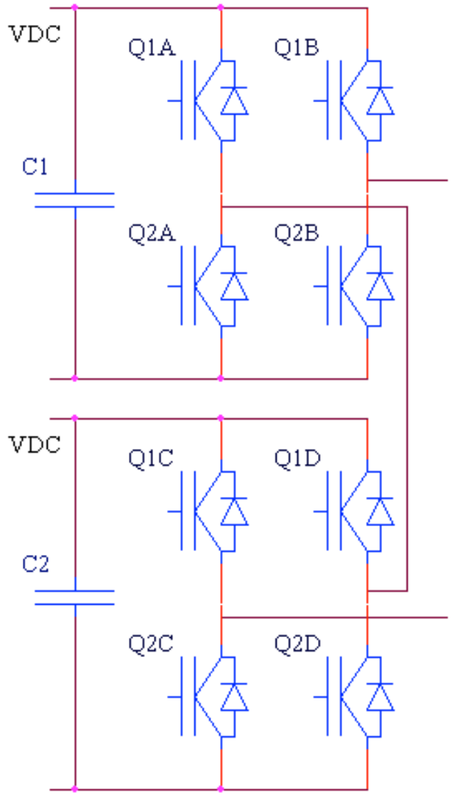
Per avere 3 possibili livelli di tensione sui rami dell'inverter, gli IGBT sono controllati in modo tale che in ogni momento solo 2 dei 4 per ogni gamba siano in conduzione.

Nel circuito a lato, quando sono in conduzione Q1A e Q2A la tensione superiore di bus (+VDC) viene applicata alla fase A, quando sono in conduzione Q2A e Q3A gli viene applicata la tensione del punto neutro comune (zero volt), mentre quando sono in conduzione Q3A e Q4A in uscita della fase A si trova la tensione inferiore del bus DC (-VDC).

Conversione multilivello



3-phase clamped capacitor DC/AC converter



Inverter monofase (o gamba di un inverter trifase) a 5-livelli realizzato con la cascata di 2 inverter a 2-livelli

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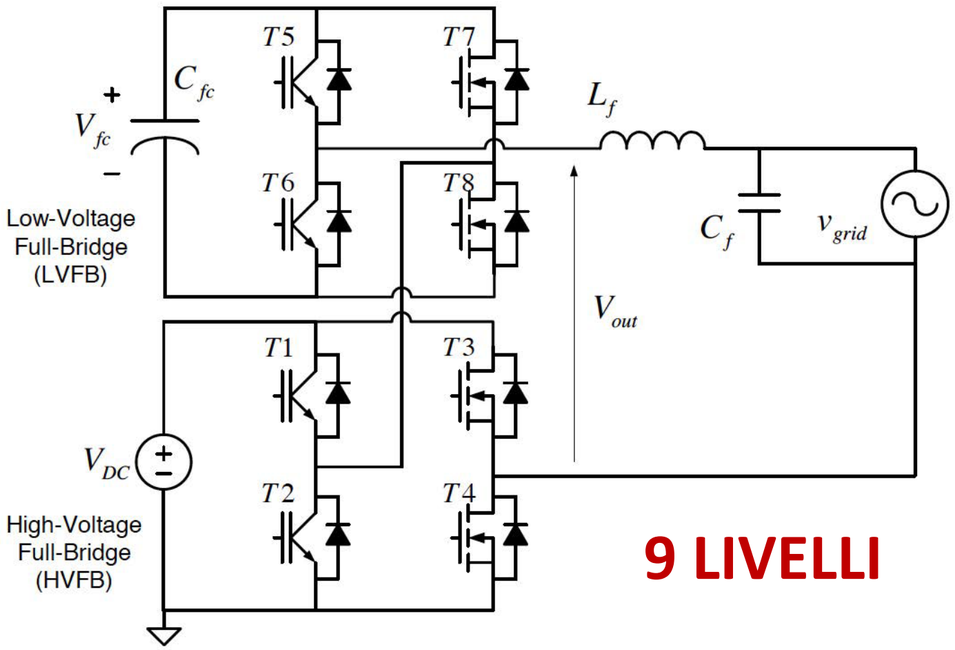
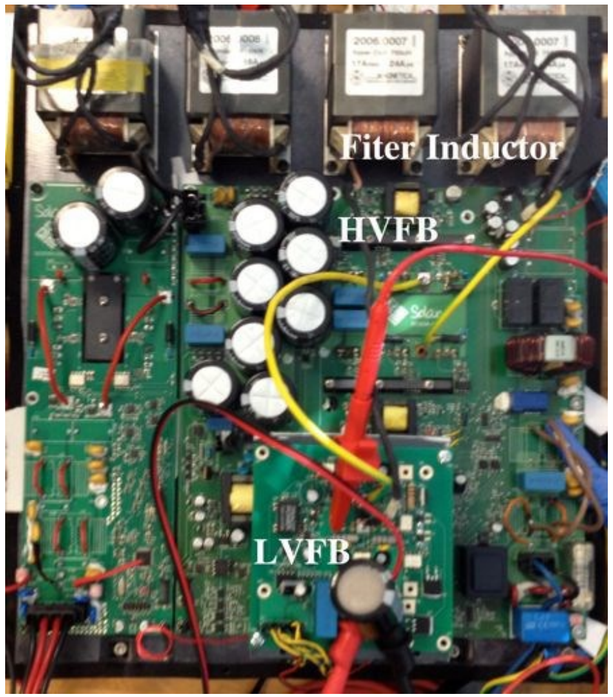
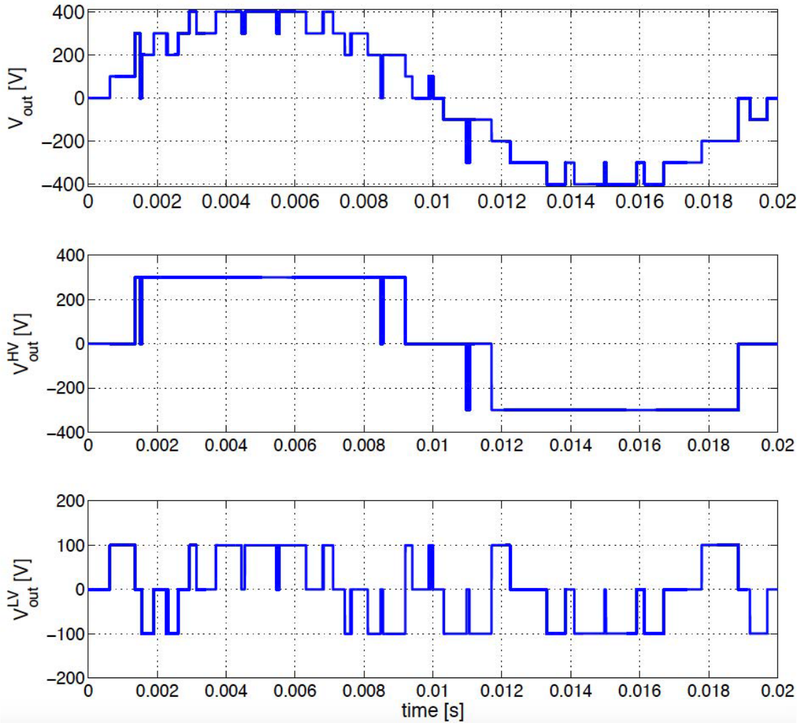
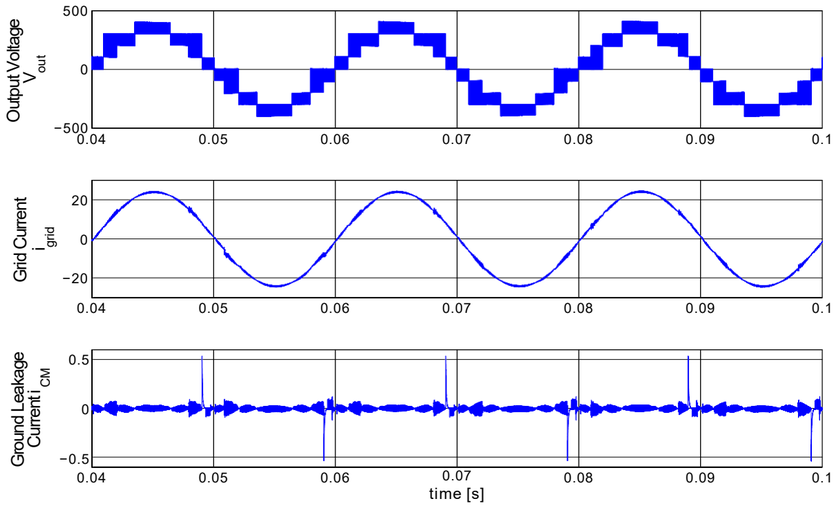
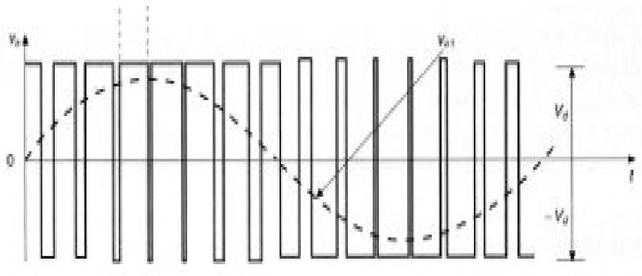


9 LIVELLI

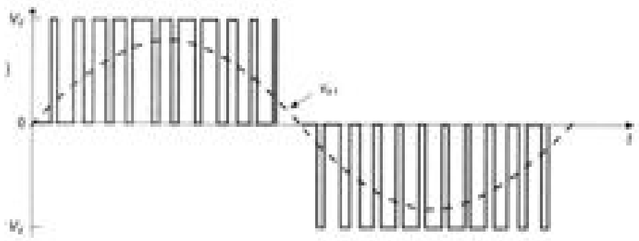








2 livelli



3 livelli